

TASK 10

Calculate the message digest of a text using the MD5 algorithm in JAVA.

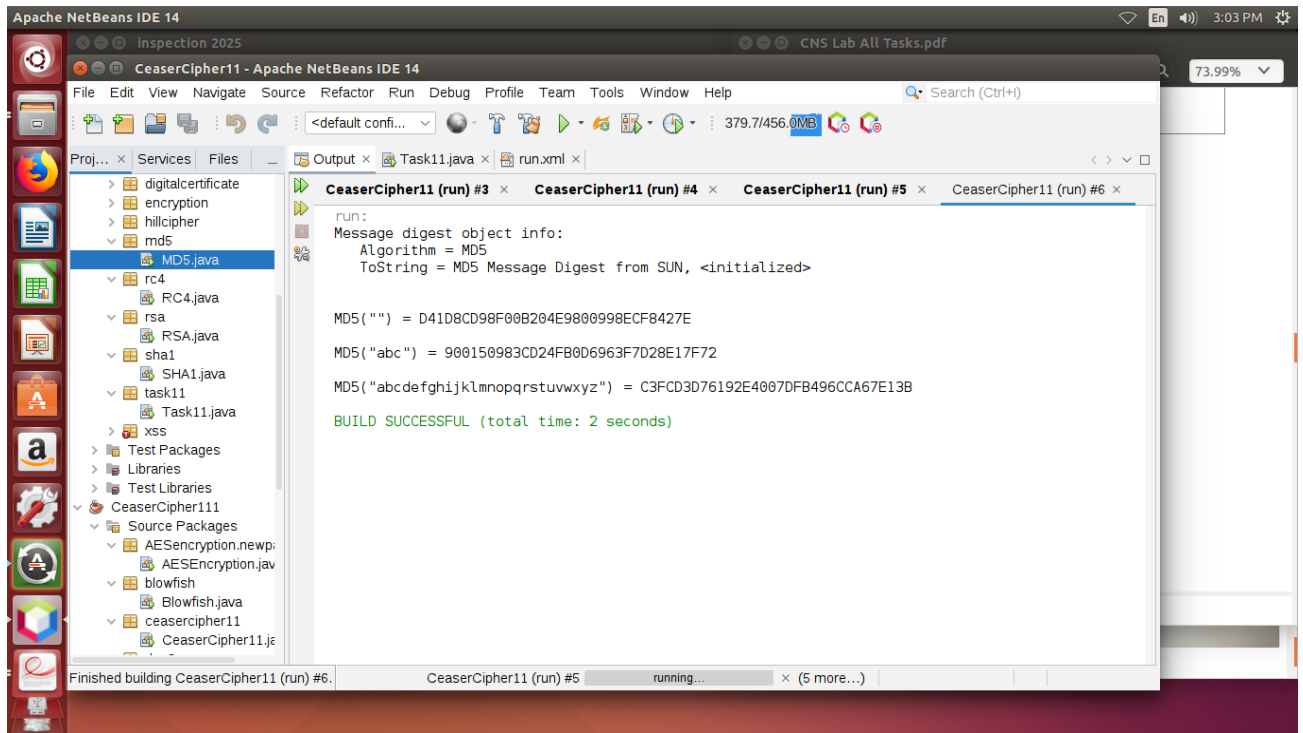
Aim: To calculate the message digest of a text using the MD5 algorithm in JAVA.

Program:

```
import java.security.*;
import java.util.*;
public class Task10 {
    static Scanner s = new
Scanner(System.in); public static void
main(String[] args) {
    try {
        System.out.print("Enterinputdata:");
        String data = s.nextLine();

        MessageDigest md = MessageDigest.getInstance("MD5");
        System.out.println("Message digest object info: ");
        System.out.println(" Algorithm = " +
md.getAlgorithm());
        System.out.println("ToString="+
md.toString());md.update(data.getBytes());
        System.out.println("MD5(\"" + data + "\")="+HexFormat.of().formatHex(md.digest()));
    }
    catch (Exception e) {
        e.printStackTrace();
    }
}
}
```

OUTPUT:



```
run:
Message digest object info:
  Algorithm = MD5
  ToString = MD5 Message Digest from SUN, <initialized>

MD5("") = D41D8CD98F00B204E9800998ECF8427E
MD5("abc") = 900150983CD24FB0D6963F7D28E17F72
MD5("abcdefghijklmnopqrstuvwxyz") = C3FCD3D76192E4007DFB496CCA67E13B

BUILD SUCCESSFUL (total time: 2 seconds)
```

VIVA QUESTIONS

1. How many rounds does MD5 use in its hashing process?
2. Explain the step-by-step working of the MD5 algorithm.
3. What are the four main operations used in MD5's compression function?
4. What is the length of the MD5 hash output?